Data Warehouses

Letizia Tanca (*)
Politecnico di Milano

(*) with the kind support of Rosalba Rossato

Why Data Warehouses

- Data should be integrated across the enterprise(s)
- Historical data hold the key to understanding data over time
- Summary data provide an important value to the organization
- What-if capabilities are required

Outline

- What is a Data Warehouse
- Data Warehouse Architecture
- Data Warehouse Design

What is a Data Warehouse?

A data warehouse is a single, complete and consistent **store of data**, obtained from a variety of different sources and made available to end users so that they can understand and use it in a business context.

[Barry Devlin]

A data warehouse provides techniques for assembling and managing data from various sources, with the purpose of answering business questions and thus making decisions that were not possible previously

Where is a DW useful: examples

- Commerce: sales and complaints analysis, client fidelization, shipping and stock control
- Manufacturing plants: production cost control, provision and order support
- Financial services: risk and credit card analysis, fraud detection
- Telecommunications: call flow analysis, subscribers' profiles
- Healthcare structures: patients' ingoing and outgoing flows, cost analysis
- ……….

Examples of DW queries

- Show total sales across all products at increasing aggregation of geographic levels, from state, to country, to geographical region (e.g., Connecticut-USA-Americas), for 1999 and 2000.
- Create a cross-tabular analysis of our operations, showing expenses by territory in South America for 1999 and 2000. Include all possible subtotals.
- List the top 10 sales representatives of automotive products in Asia according to sales revenue in year 2000, and rank their commissions.

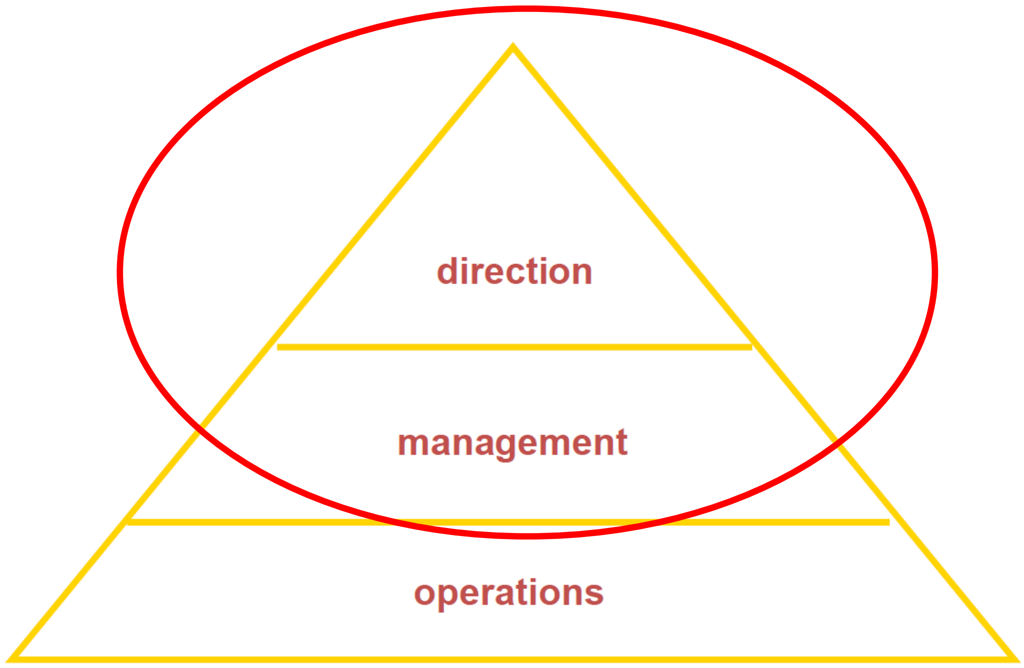
An operational definition

A Data Warehouse is a

- subject-oriented,
- integrated,
- time-varying,
- non-volatile

collection of data that is used primarily in **organizational decision making**.

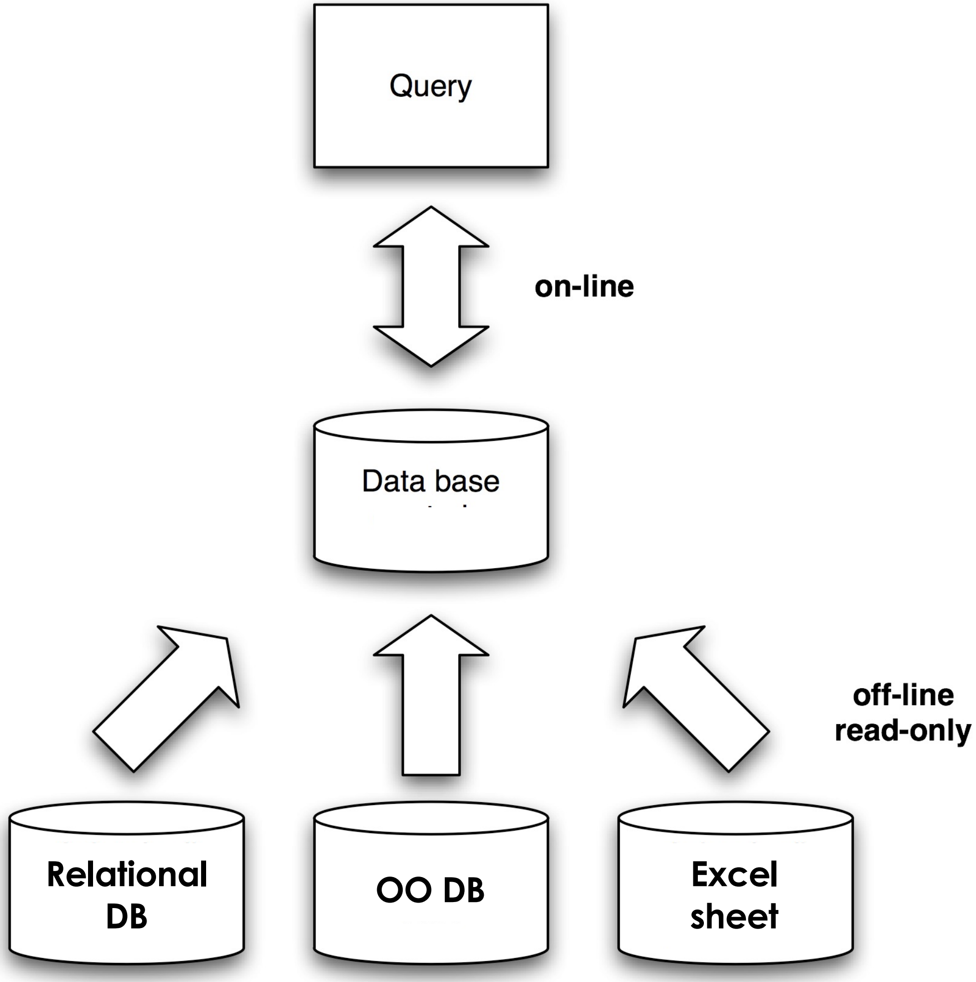
[Bill Inmon, Building the Data Warehouse, 1996]



Recall:
Materialized Data integration

- The main purpose of a data warehouse is to allow systematic or ad-hoc data analysis and mining.

- In a dynamic environment, one must performthe so-called ETL (Extract, Transform, Load) periodically – say, once a day or a week or a month-, thereby building up a history of the enterprise.



Dimensions of a Data Warehouse

Data warehouses are very large databases

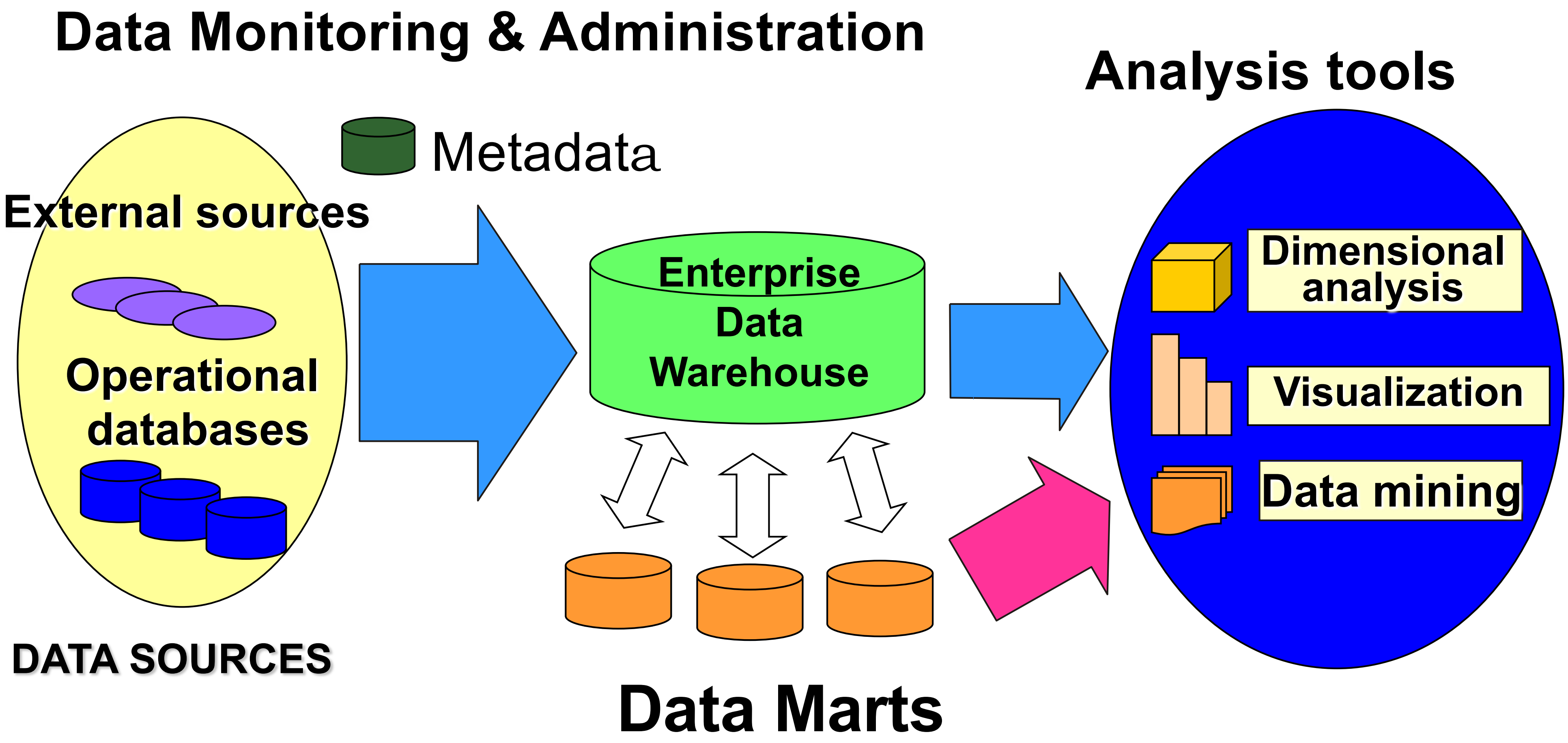
- Terabytes (10^{12} bytes): most small or medium-sized organizations have data warehouse sizes in the range 1 to 5 terabytes, though some have to deal with larger ones.
- Petabytes (10^{15} bytes): e.g. Geographic Information Systems (GIS)
- Exabytes (10^{18} bytes): e.g. National Medical Records
- Zettabytes (10^{21} bytes): e.g. Weather reports, including images
- Zottabytes (10^{24} bytes): e.g. Intelligence Agency Videos
- ……….

DW vs Data Lakes - aka Dataspaces

(we will introduce and discuss this subject again in later lectures)

Data Warehouse	Dataspace
Usually Relational DBMS	Often HDFS/Hadoop
Data of recognized high value	Candidate data of potential value
Aggregated data	Detailed source data
Data conforms to enterprise standards	Fidelity to original format and condition (transformed/cleaned later, on demand, when needed)
Known entities	Raw material (metadata to discover entities)
Data integration up front	Data integration on demand
Schema on write	Schema on read (when doing analytics)

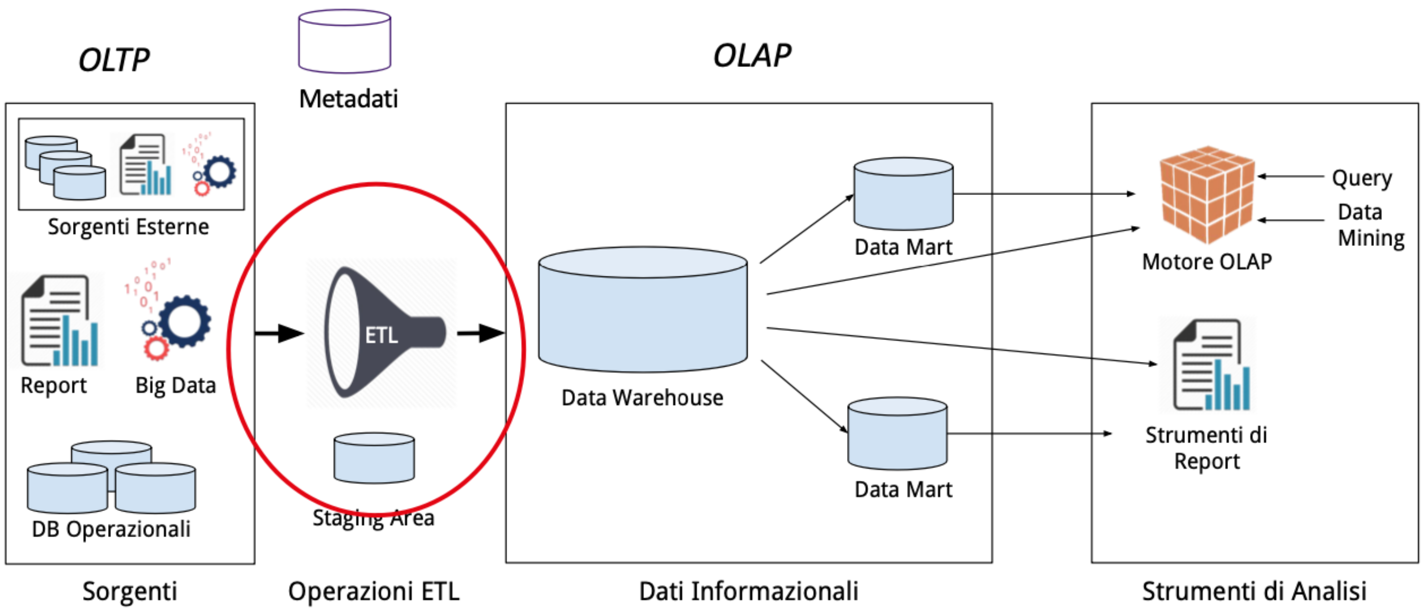
Architecture for a data warehouse



Data Quality is Fundamental also for DWs

(Recall prof. Cappiello's slide)

Data Quality improvement methods are also used in Data Warehouse



Evolutions of Data Warehouses

- **Offline data warehouse:** <u>periodically updated</u> from the data of the operational systems, and the DW data are stored in a data structure designed to facilitate reporting.
- **Online data warehouse:** data in the warehouse is <u>kept updated for every transaction</u> performed on the source data (e.g. by triggers)

- **Single-source data warehouse:** data extracted from a unique data source
- **Integrated data warehouse:** data assembled from different data sources, so users can look up the information they need across other systems

Can DWs be replaced by Big Data infrastructures?

- New Big Data Technologies, such as cloud-based data stores, cannot just replace Data Warehouses
- Data Warehouse is a paradigm, a model, therefore a DW can surely be stored into a cloud-based repository
- The two technologies are thus complementary

Normally a DW is stored in a relational DBMS

……. but it is a specialized DB, where we apply a different kind of operations w.r.t. those of a transactional one

Standard (Transactional) DB:
(OLTP = OnLine Transactional Processing)

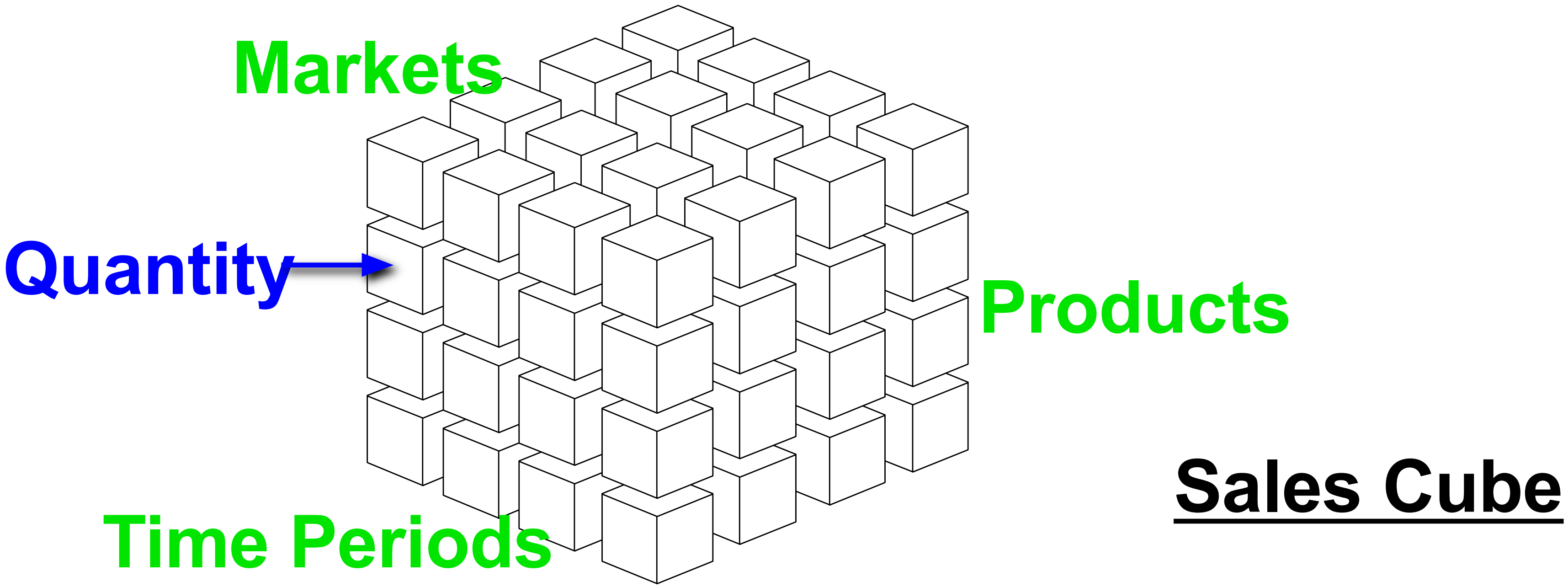
- Mostly updates
- Many small transactions
- Gb – Pb (10^9- 10^{15} bytes) of data
- Current snapshot
- Index/hash on p.k.
- Raw data
- Thousands of users (typically, generic employees)

DataWarehouse:
(OLAP = OnLine Analytical Processing)

- Mostly reads
- Queries are long and complex
- Pb – Eb (10^{15}- 10^{18} bytes) of data
- History
- Lots of scans
- Summarized, reconciled data
- Hundreds of users (typically, company management)

OLAP-oriented data models

- We want to study the data from different viewpoints (dimensions) and at different levels of granularity (hierarchies)
- Most appropriate representation: the data cube metaphor



OLAP-oriented data models

- Must support sophisticated analyses and computations over different *dimensions* and *hierarchies*
- Most dimensions are hierarchical, e.g.:
  - DATE {DAY-MONTH-TRIMESTER-YEAR}
  - PRODUCT {BRAND - TYPE - CATEGORY}

    (e.g. LAND ROVER - CARS - VEHICLES)
- The cube cells contain *metric values*
- Usually the Data Cube model is implemented over a relational DBMS – and we will study this approach

Dimensional Fact Model (DFM)

- Allows one to describe a set of

  fact schemata
- The components of a fact schema are:
  - Facts
  - Measures
  - Dimensions
  - Dimension Hierarchy

DFM: elements

- A fact is a concept that is relevant for the decisional process; typically, it models a set of events of the organization (e.g., sales)
- A measure is a numerical property of a fact (e.g. nr. of sales)
- A dimension is a fact property defined w.r.t. a finite domain; it describes an analysis coordinate for the fact (e.g.,the sales of which product)

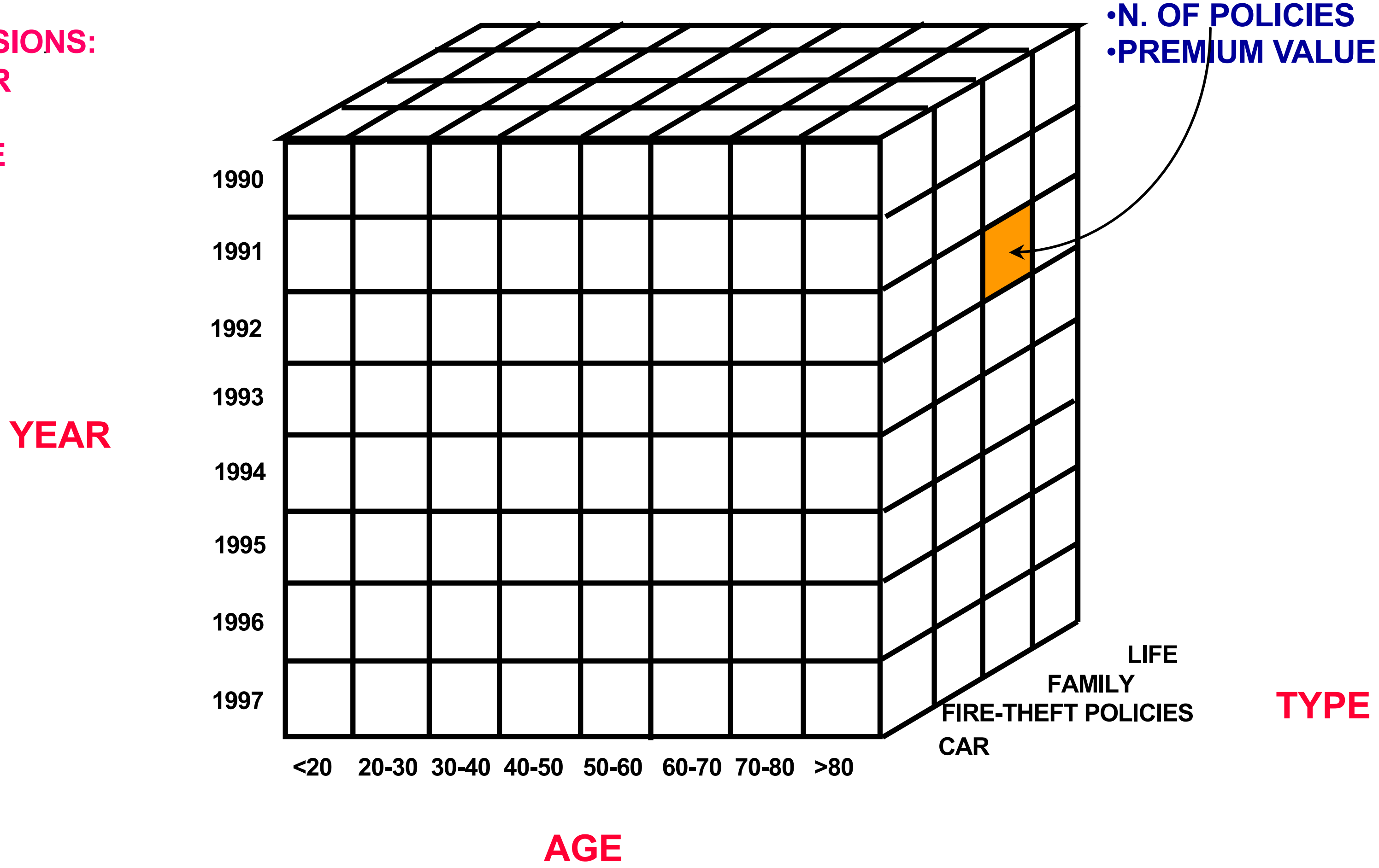
LOGICAL MODELS FOR OLAP

EXAMPLE :THE INSURANCE COMPANY DATA CUBE

FACT: POLICIES

DIMENSIONS:
- YEAR
- AGE
- TYPE

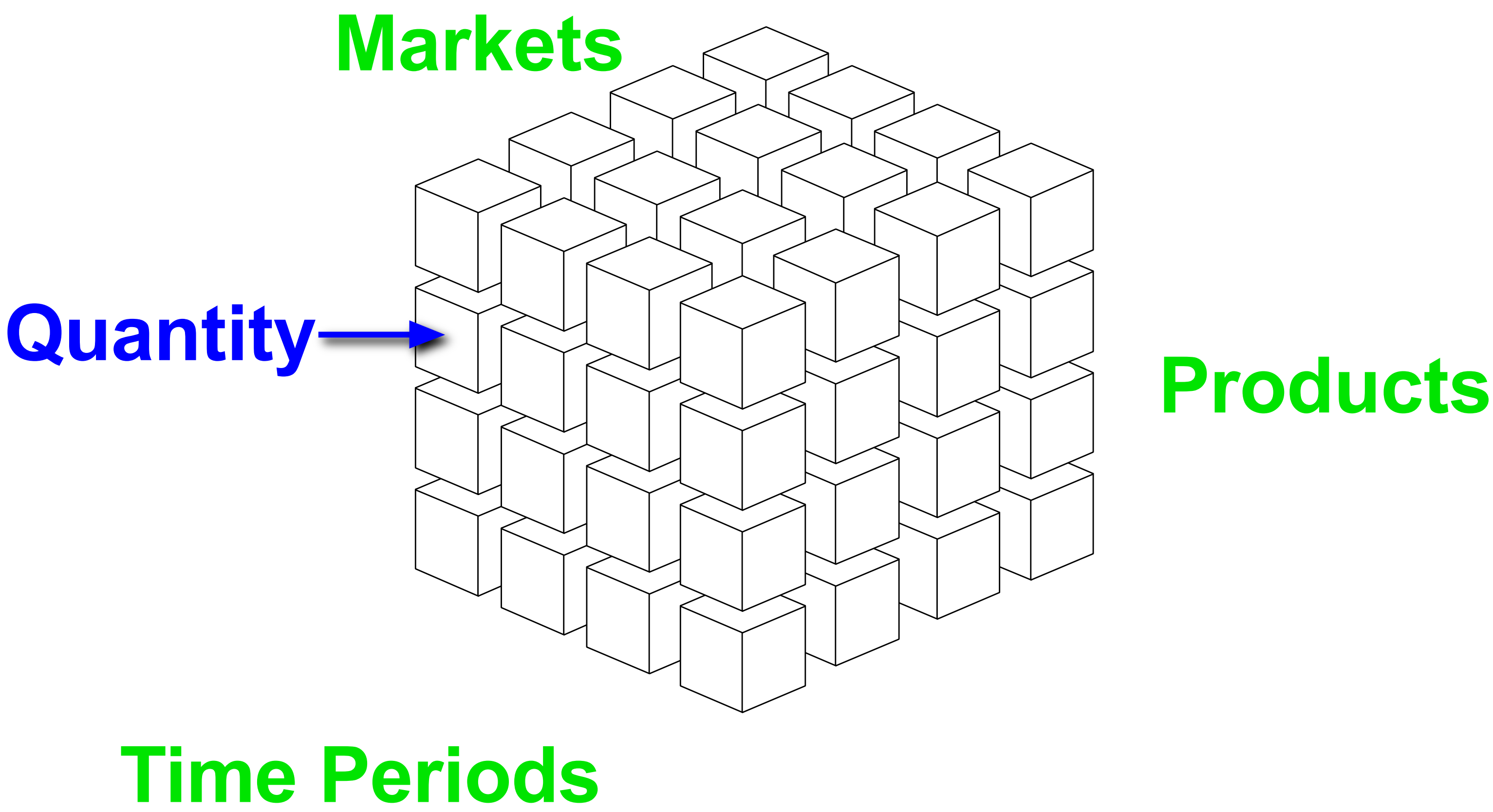
METRICS:
- •N. OF POLICIES
- •PREMIUM VALUE



Other examples

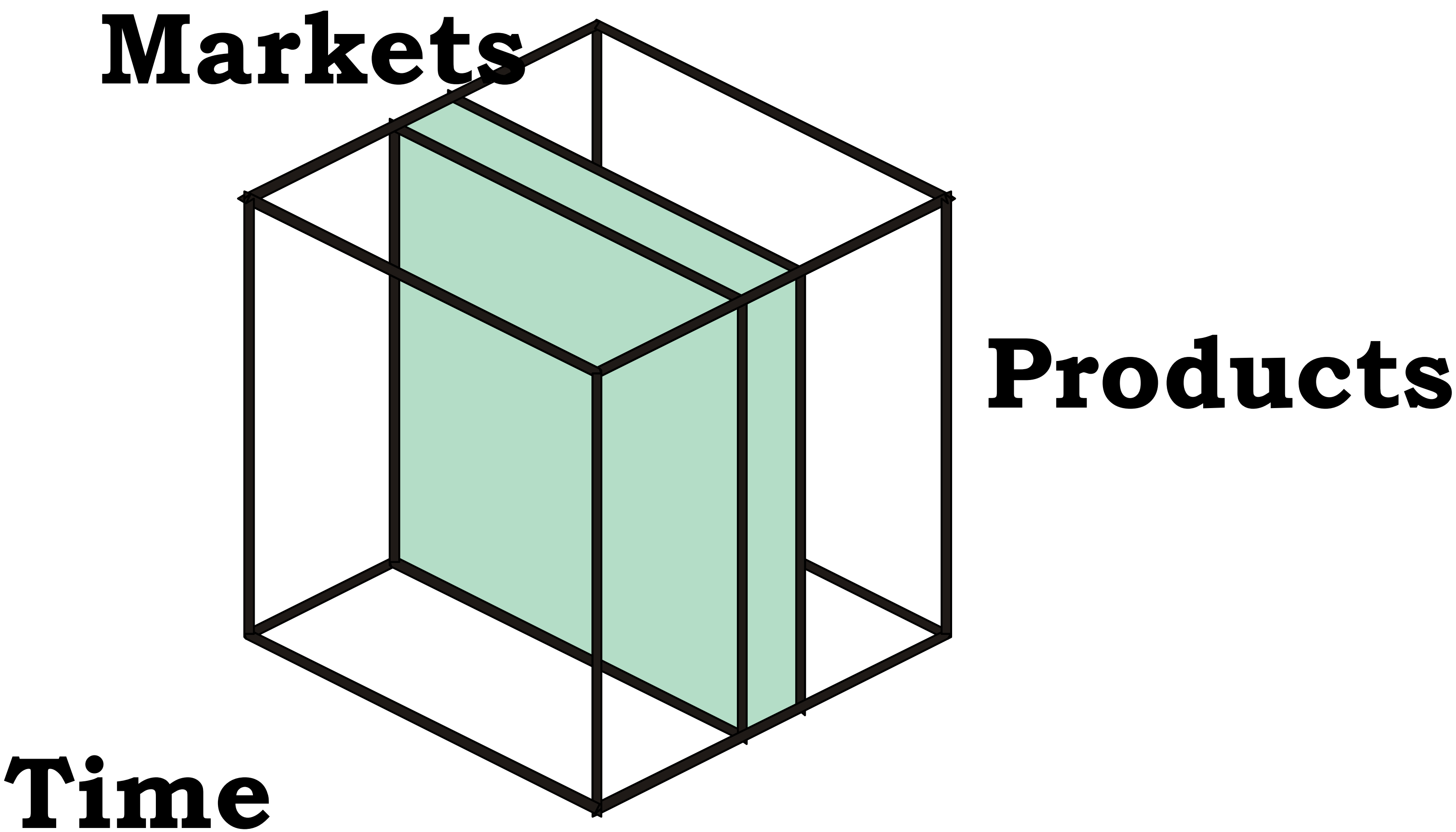
- Store chain
  - Fact: sales
  - Measures: sold quantity, gross income
  - Dimensions: product, time, zone
- Telecom Operator
  - Fact: phone call
  - Measures : cost, duration
  - Dimensions: caller subscriber,called subscriber, time

Multidimensional Representation of the Different Views

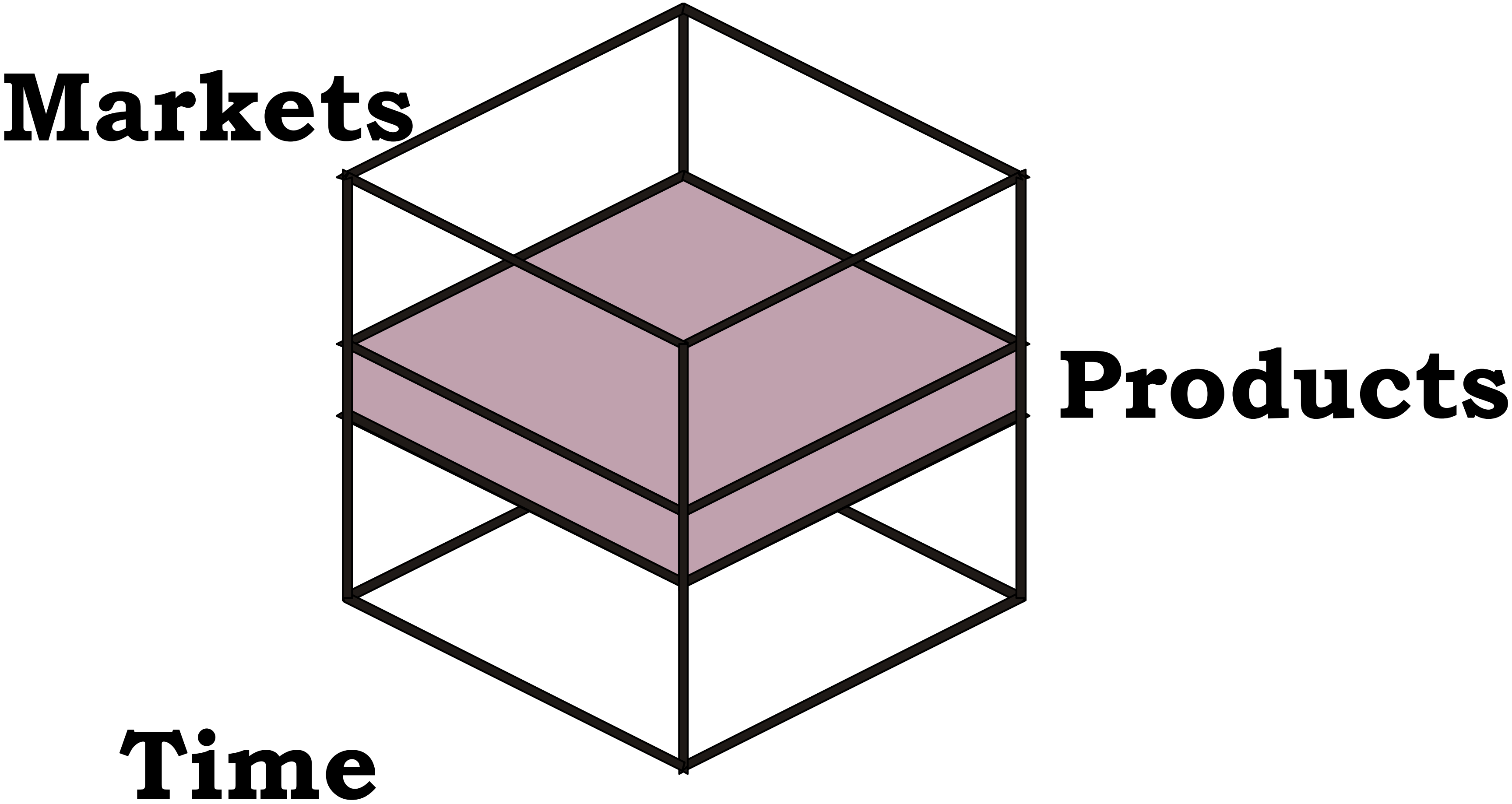


Sales Cube

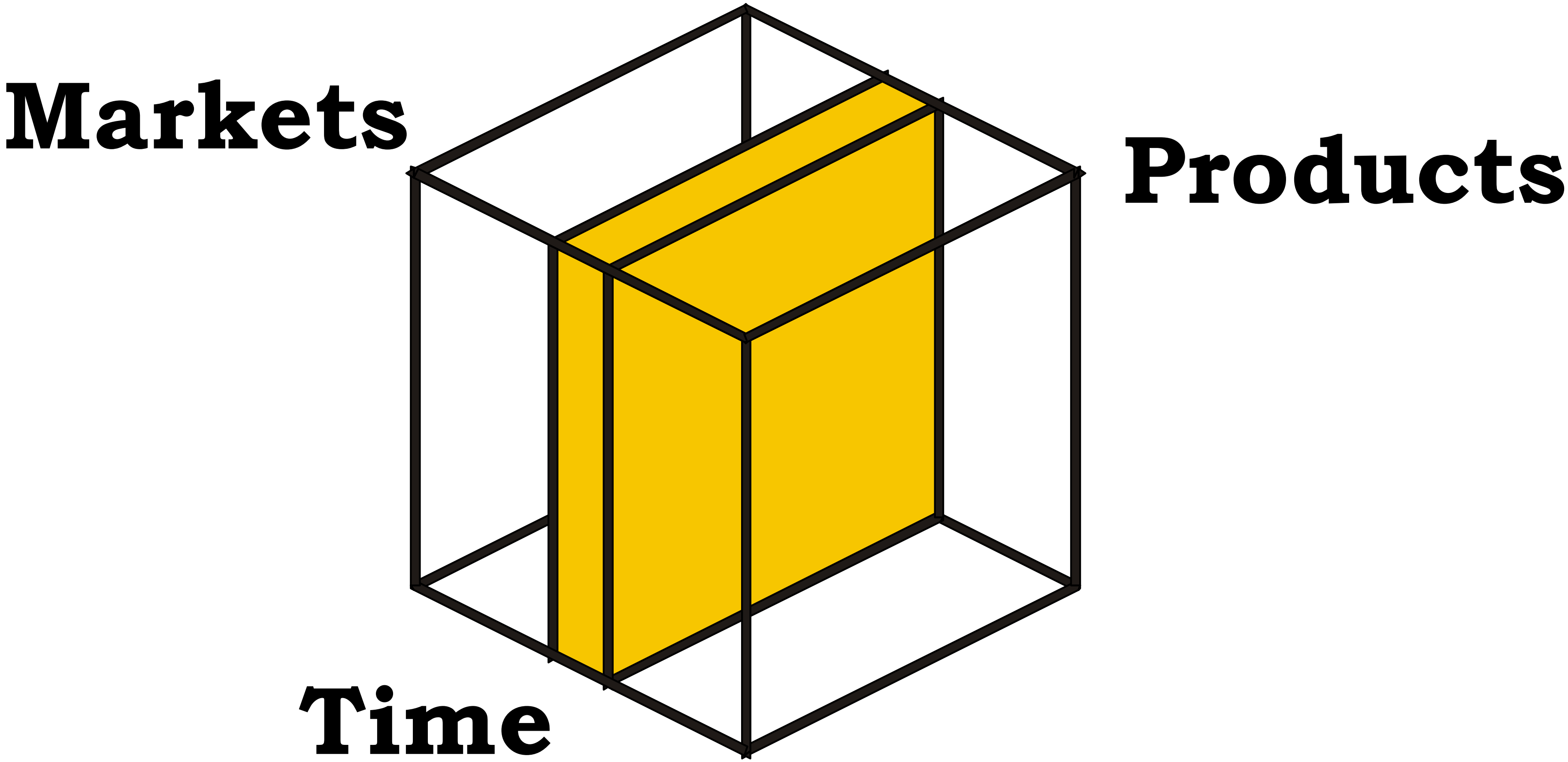
The area manager examines product sales of his/her own markets in all periods and for all products



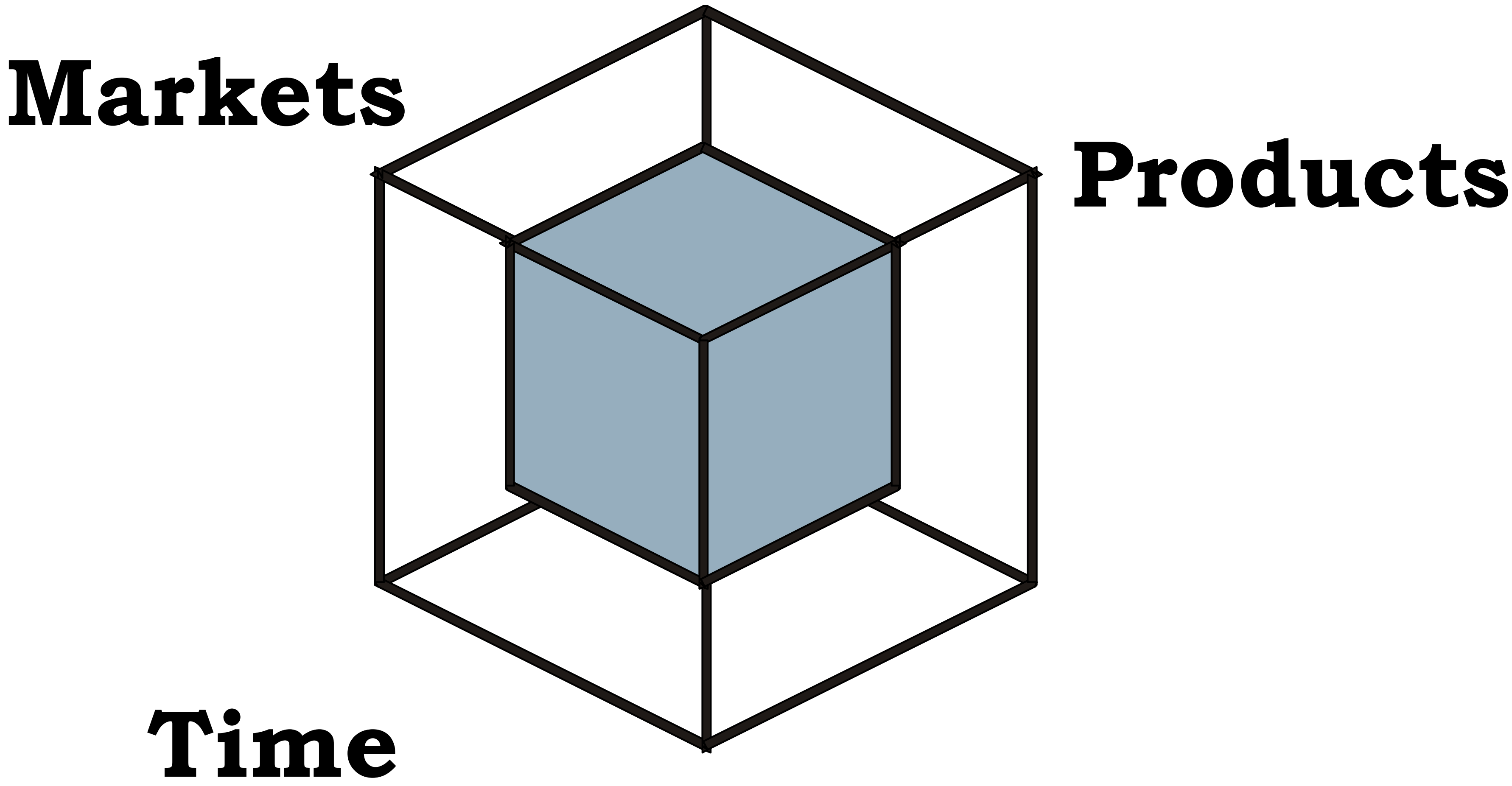
The product manager examines the sales of some specific products in all periods and in all markets



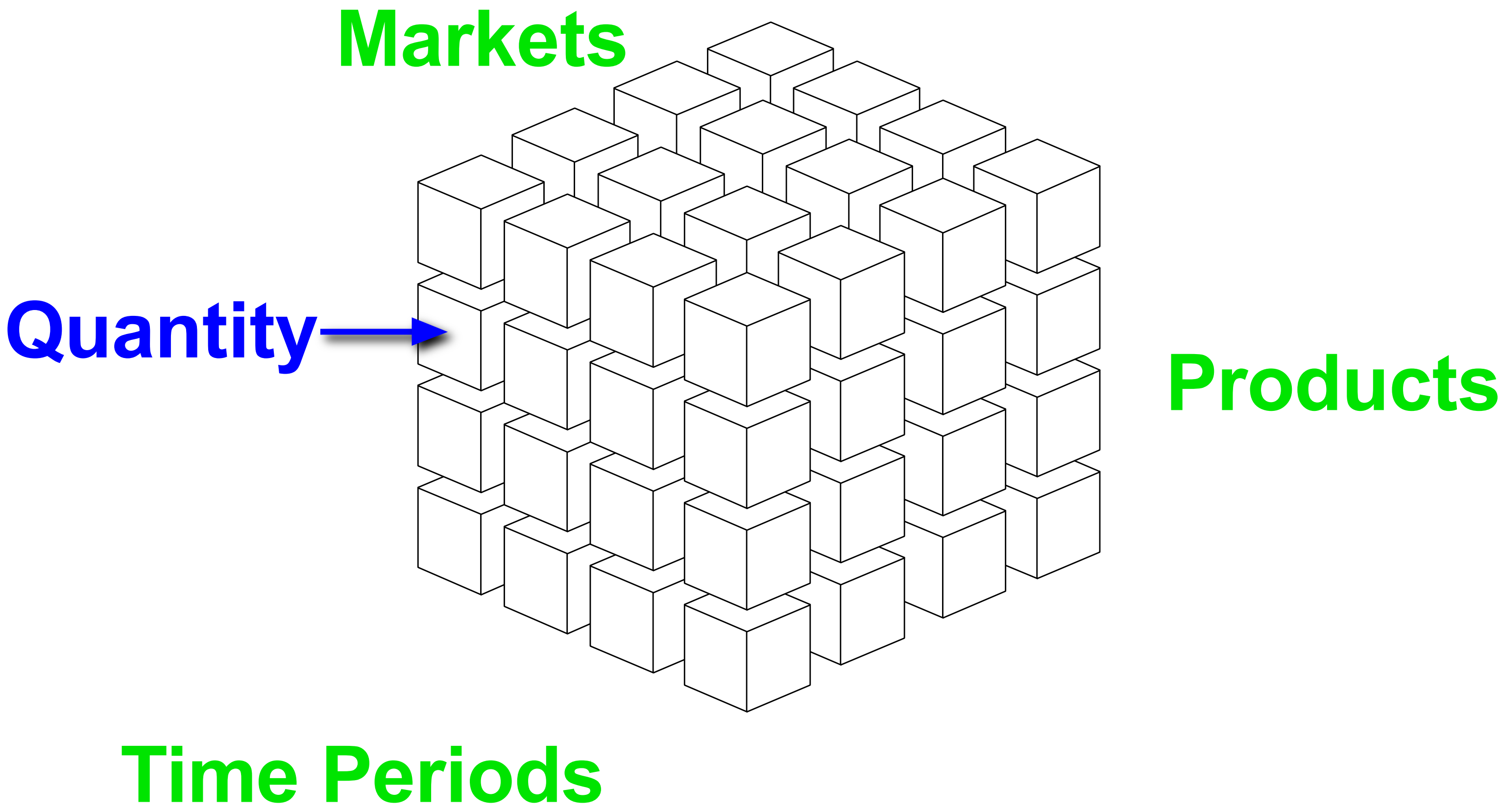
The financial manager examines product sales in all markets, for a given time period



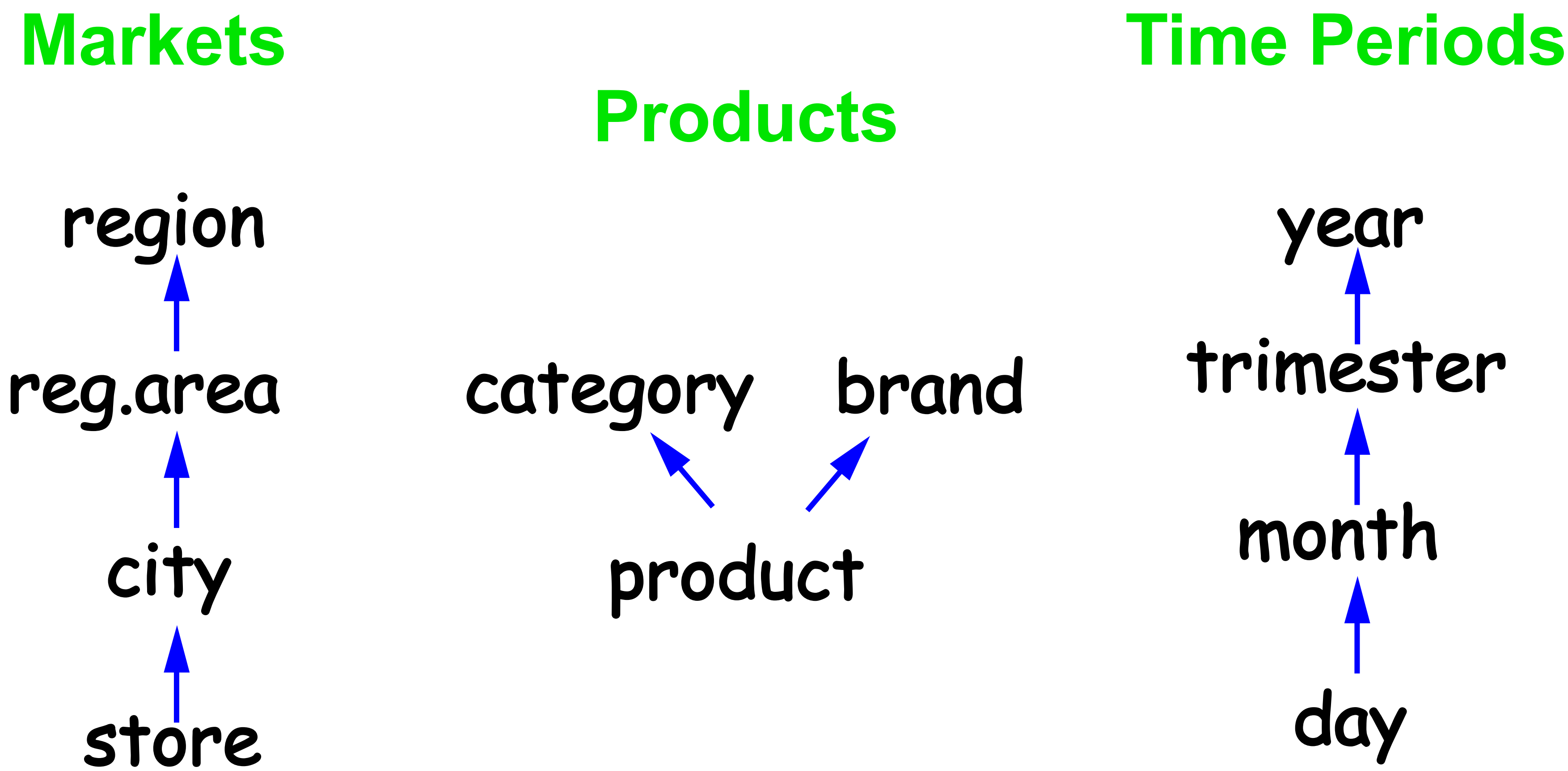
The strategic manager concentrates on a category of products, a specific region and a medium time span



Multidimensional Representation: Possible Hierarchies of the Sales Cube



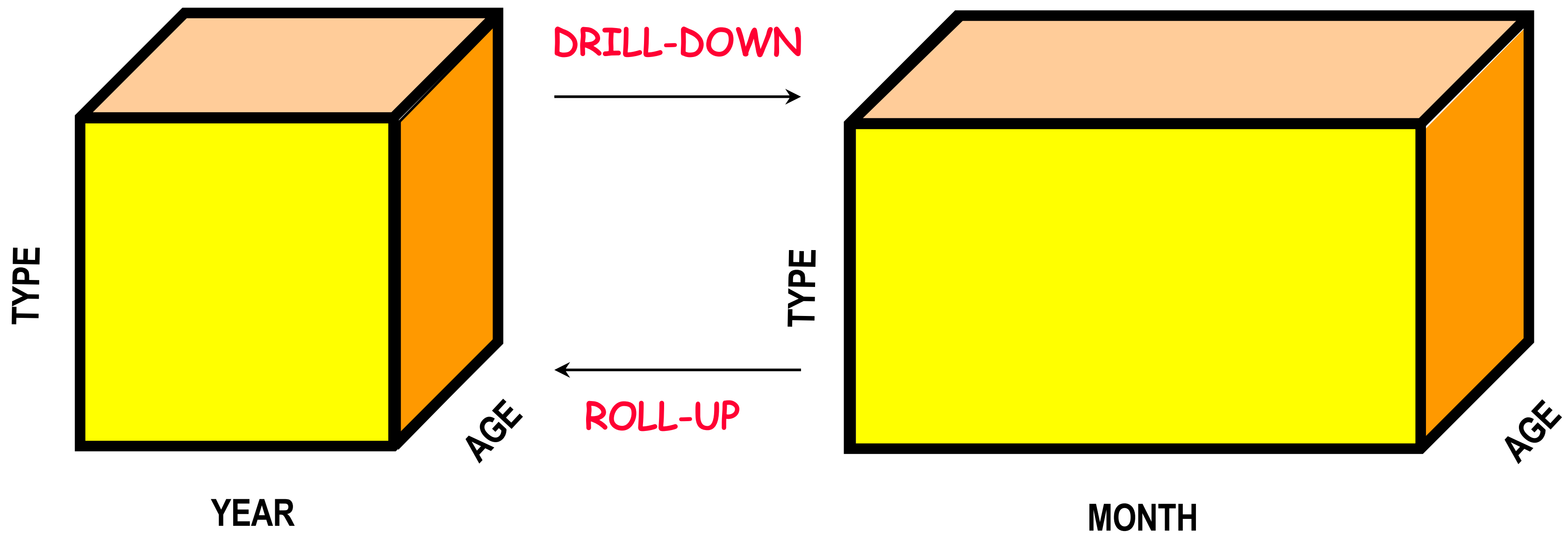
Dimensions and hierarchies



OLAP OPERATIONS

- roll-up
  - Aggregates data at a higher level – **e.g. last year's sales volume per product category and per region**
- drill-down
  - De-aggregates data at the lower level – **e.g. for a given product category and a given region, show daily sales**
- slice-and-dice
  - Applies selections and projections, **which reduce data dimensionality**
- pivoting
  - Selects two dimensions to re-aggregate data (**cube re-orientation**)
- ranking
  - **Sorts data** according to predefined criteria
- traditional operations (select, project, join, derived attributes, etc.)

OLAP OPERATIONS



Roll-up (2 dimensions)

Metrics	Dollar Sales									
Customer Region	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany
Month										
Jan 97	$ 620	$ 753	$ 30	$ 660	$ 2.405	$ 1.312	$ 440	$ 1.002	$ 1.002	$ 383
Feb 97	$ 258	$ 252	$ 800	$ 975	$ 160	$ 582	$ 744	$ 310	$ 799	$ 118
Mar 97	$ 648	$ 244	$ 148	$ 250	$ 1.085	$ 2.961	$ 650	$ 1.240	$ 119	$ 142
Apr 97	$ 787	$ 588	$ 447	$ 486	$ 226	$ 506	$ 601	$ 119	$ 550	$ 85
May 97	$ 1.350	$ 245	$ 936	$ 159	$ 664	$ 626	$ 107	$ 135	$ 200	$ 177
Jun 97	$ 842	$ 582	$ 1.281	$ 937	$ 240	$ 774	$ 176	$ 1.139	$ 652	$ 254
Jul 97	$ 652	$ 690	$ 486	$ 1.293	$ 605	$ 303	$ 818	$ 103	$ 124	$ 173
Aug 97	$ 1.783	$ 304	$ 1.032	$ 170	$ 398	$ 356	$ 432	$ 190	$ 241	$ 407
Sep 97	$ 581	$ 778	$ 3.558	$ 587	$ 440	$ 1.652	$ 1.071	$ 315	$ 210	$ 202
Oct 97	$ 2.291	$ 1.840	$ 600	$ 656	$ 1.300	$ 718	$ 1.210	$ 427	$ 220	$ 520
Nov 97	$ 39	$ 1.602	$ 1.082	$ 1.187	$ 842	$ 759	$ 745	$ 232	$ 101	$ 1.037
Dec 97	$ 381	$ 1.588	$ 343	$ 118	$ 1.459	$ 635	$ 2.021	$ 259	$ 210	$ 119
Jan 98	$ 311	$ 1.174	$ 2.634	$ 3.130	$ 954	$ 2.083	$ 1.351	$ 747	$ 426	$ 447
Feb 98	$ 2.518	$ 702	$ 1.123	$ 1.336	$ 1.227	$ 3.887	$ 545	$ 268	$ 277	$ 282
Mar 98	$ 2.459	$ 1.523	$ 1.178	$ 4.708	$ 1.420	$ 3.514	$ 1.948	$ 1.705	$ 276	$ 1.168
Apr 98	$ 407	$ 841	$ 524	$ 712	$ 133	$ 2.486	$ 49	$ 390	$ 1.298	$ 221
May 98	$ 667	$ 1.721	$ 440	$ 148	$ 80	$ 1.310	$ 303	$ 104	$ 657	$ 65
Jun 98	$ 699	$ 1.096	$ 898	$ 353	$ 902	$ 839		$ 230	$ 155	$ 105
Jul 98	$ 586	$ 1.897	$ 412	$ 226	$ 406	$ 361	$ 1.628	$ 267	$ 1.011	$ 41
Aug 98	$ 894	$ 326	$ 792	$ 1.832	$ 1.199	$ 295	$ 1.816	$ 277	$ 102	$ 118
Sep 98	$ 338	$ 3.179	$ 505	$ 427	$ 99	$ 2.976	$ 885	$ 135	$ 85	$ 1.110
Oct 98	$ 544	$ 413	$ 1.467	$ 209	$ 679	$ 706	$ 556	$ 480	$ 485	$ 99
Nov 98	$ 671	$ 459	$ 1.471	$ 2.066	$ 701	$ 716	$ 986	$ 1.127	$ 154	$ 440
Dec 98	$ 836	$ 2.096	$ 1.726	$ 3.642	$ 395	$ 1.740	$ 1.943	$ 1.143	$ 366	$ 307

Metrics	Dollar Sales									
Customer Region	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France	Germany
Quarter										
Q1 1997	$ 1.526	$ 1.249	$ 978	$ 1.885	$ 3.650	$ 4.855	$ 1.834	$ 2.552	$ 1.920	$ 643
Q2 1997	$ 2.979	$ 1.415	$ 2.664	$ 1.582	$ 1.130	$ 1.906	$ 884	$ 1.393	$ 1.402	$ 516
Q3 1997	$ 3.016	$ 1.772	$ 5.076	$ 2.050	$ 1.443	$ 2.311	$ 2.321	$ 608	$ 575	$ 782
Q4 1997	$ 2.711	$ 5.030	$ 2.025	$ 1.961	$ 3.601	$ 2.112	$ 3.976	$ 918	$ 531	$ 1.676
Q1 1998	$ 5.288	$ 3.399	$ 4.935	$ 9.174	$ 3.601	$ 9.484	$ 3.844	$ 2.720	$ 979	$ 1.897
Q2 1998	$ 1.773	$ 3.658	$ 1.862	$ 1.213	$ 1.115	$ 4.635	$ 352	$ 724	$ 2.110	$ 391
Q3 1998	$ 1.818	$ 5.402	$ 1.709	$ 2.485	$ 1.704	$ 3.632	$ 4.329	$ 679	$ 1.198	$ 1.269
Q4 1998	$ 2.051	$ 2.968	$ 4.664	$ 5.917	$ 1.775	$ 3.162	$ 3.485	$ 2.750	$ 1.005	$ 846

Roll-up (3 dimensions)

	Metrics	Dollar Sales								
	Customer Region	North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England	France
Category	Year									
Electronics	1997	$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566	$ 199
	1998	$ 1.184	$ 4.529	$ 1892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683	$ 462
Food	1997	$ 759	$ 682	$ 729	$ 262	$ 588	$ 469	$ 807	$ 156	$ 615
	1998	$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165	$ 175
Gifts	1997	$ 2.532	$ 1.355	$ 1854	$ 1.413	$ 2.535	$ 2.132	$ 1.904	$ 908	$ 375
	1998	$ 1.955	$ 2.785	$ 2800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158	$ 717
Health & Beauty	1997	$ 624	$ 640	$ 1317	$ 647	$ 588	$ 754	$ 654	$ 143	$ 292
	1998	$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273	$ 72
Household	1997	$ 5.354	$ 4.112	$ 5410	$ 4.446	$ 3.058	$ 3.974	$ 2.654	$ 3.545	$ 2.875
	1998	$ 5.787	$ 5.320	$ 5416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139	$ 3.649
Kid's Korner	1997	$ 201	$ 398	$ 485	$ 186	$ 409	$ 323	$ 306	$ 105	$ 34
	1998	$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198	$ 19
Travel	1997	$ 624	$ 505	$ 564	$ 386	$ 300	$ 978	$ 416	$ 48	$ 38
	1998	$ 608	$ 559	$ 1096	$ 611	$ 464	$ 316	$ 573	$ 257	$ 198

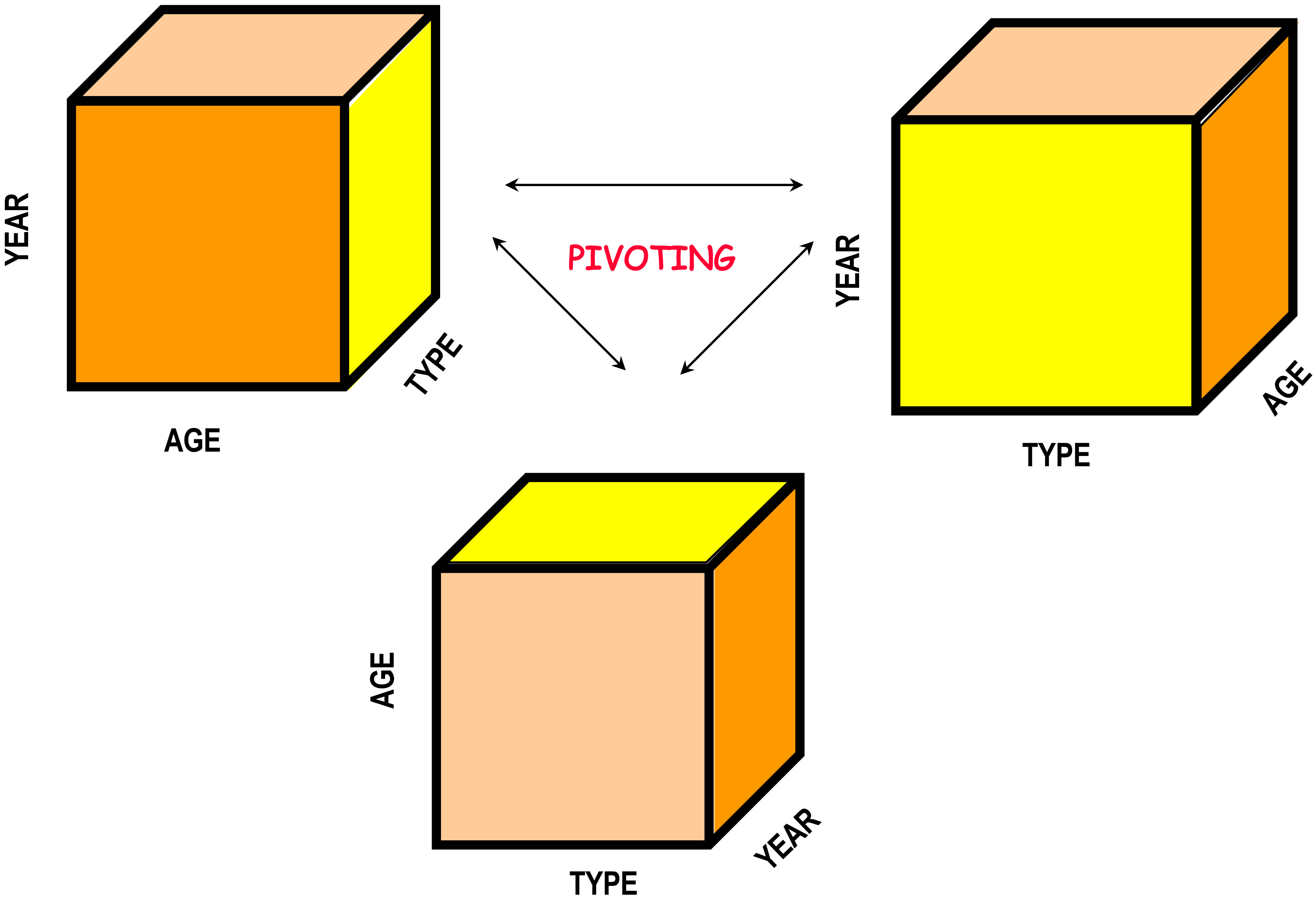
	Metrics	Dollar Sales
Category	$\boxed{\text{Year}}$	
Electronics	1997	$ 10.616
	1998	$ 29.299
Food	1997	$ 5.300
	1998	$ 5.638
Gifts	1997	$ 16.315
	1998	$ 20.047
Health & Beauty	1997	$ 6.042
	1998	$ 5.665
Household	1997	$ 38.383
	1998	$ 50.391
Kid's Korner	1997	$ 2.559
	1998	$ 2.943
Travel	1997	$ 4.497
	1998	$ 4.792

Drill-down

Metrics	Dollar Sales	
Year	1997	1998
Category		
Electronics	$ 10.616	$ 29.299
Food	$ 5.300	$ 5.638
Gifts	$ 16.315	$ 20.047
Health & Beauty	$ 6.042	$ 5.665
Household	$ 38.383	$ 50.391
Kid's Korner	$ 2.559	$ 2.943
Travel	$ 4.497	$ 4.792

Metrics	Dollar Sales											
Customer Region	North-East		Mid-Atlantic		South-East		Central		South		North-West	
Year	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998	1997	1998
Category												
Electronics	$ 138	$ 1.184	$ 1.774	$ 4.529	$ 384	$ 1.892	$ 138	$ 7.232	$ 2.346	$ 651	$ 2.554	$ 9.488
Food	$ 759	$ 538	$ 682	$ 925	$ 729	$ 959	$ 262	$ 677	$ 588	$ 213	$ 469	$ 1.503
Gifts	$ 2.532	$ 1.955	$ 1.355	$ 2.785	$ 1.854	$ 2.800	$ 1.413	$ 2.695	$ 2.535	$ 1.813	$ 2.132	$ 2.844
Health & Beauty	$ 624	$ 611	$ 640	$ 887	$ 1.317	$ 566	$ 647	$ 382	$ 588	$ 499	$ 754	$ 1.162
Household	$ 5.354	$ 5.787	$ 4.112	$ 5.320	$ 5.410	$ 5.416	$ 4.446	$ 6.812	$ 3.058	$ 4.334	$ 3.974	$ 5.008
Kid's Korner	$ 201	$ 247	$ 398	$ 422	$ 485	$ 441	$ 186	$ 380	$ 409	$ 221	$ 323	$ 592
Travel	$ 624	$ 608	$ 505	$ 559	$ 564	$ 1.096	$ 386	$ 611	$ 300	$ 464	$ 978	$ 316

OLAP OPERATIONS



Pivoting

Category	Year	Metrics: Dollar Sales
Electronics	1997	$ 10.616
	1998	$ 29.299
Food	1997	$ 5.300
	1998	$ 5.638
Gifts	1997	$ 16.315
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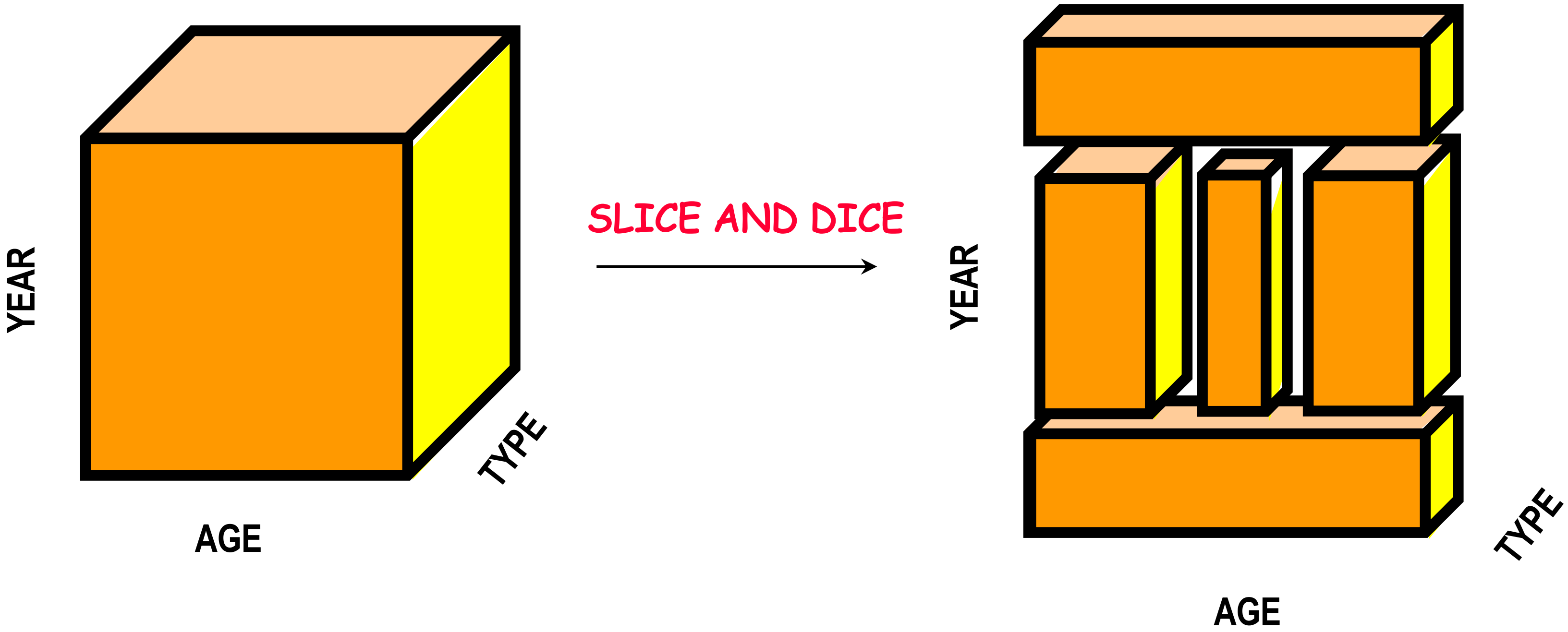

Slice and Dice

Metrics	Dollar Sales										
Subcategory / Customer City	Afton	Akron	Albon	Alcameda	Alka	Allagash	Alta	Altoola	Amestra	Amsterdam	Andersonville
Audio						$ 85					
Automotive								$ 30			
Chocolate	$ 42	$ 42		$ 50		$ 20		$ 22	$ 44		
Christmas	$ 30					$ 25	$ 30	$ 15			
Classic Toys						$ 7	$ 26				$ 38
Coffee			$ 9								
Comfort				$ 59		$ 59					
Furniture							$ 485				
Gadgets							$ 199	$ 79	$ 79		
Games & Puzzles							$ 17		$ 45		$ 45
Gift Baskets			$ 55	$ 43							
Golf	$ 25							$ 25	$ 14		$ 25
Hearth									$ 15		
Jewelry	$ 75			$ 189		$ 24	$ 77	$ 189	$ 24		
Kitchen						$ 55	$ 21		$ 76		
Lawn & Garden	$ 75		$ 100		$ 15	$ 63	$ 100		$ 180	$ 67	$ 40
Learning	$ 16							$ 37			
Meat & Cheese		$ 40		$ 20			$ 20				$ 25
Miscellaneous		$ 200	$ 1.320		$ 200	$ 139			$ 993		
Natural Remedies	$ 13								$ 13		
Pets	$ 215		$ 26			$ 30	$ 68	$ 115	$ 25		$ 34
Plants & Flowers	$ 65	$ 65	$ 65				$ 50	$ 60			
Safety & Security								$ 30	$ 22	$ 22	
Skin Care											
Sleeping			$ 18								
Toys & Accessories							$ 29	$ 185	$ 744		

Filter Details:
Category = Electronics
AND
Dollar Sales > 80
AND
Customer Region = North-West
AND
Year = 1997

Metrics	Dollar Sales					
Subcategory / Customer City	Alta	Armstrong	Avery Heights	Lane	Mt. Everest	San Fransisco
Audio		$ 98		$ 123	$ 85	
Comfort			$ 118		$ 1.495	
Gadgets	$ 199					$ 199

Slice and Dice

Category	Year	Dollar Sales: North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England
Electronics	1997	$ 138	$ 1.774	$ 384	$ 138	$ 2.346	$ 2.554	$ 2.184	$ 566
	1998	$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 476	$ 2.683
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Metrics: Dollar Sales; Customer Region across columns.

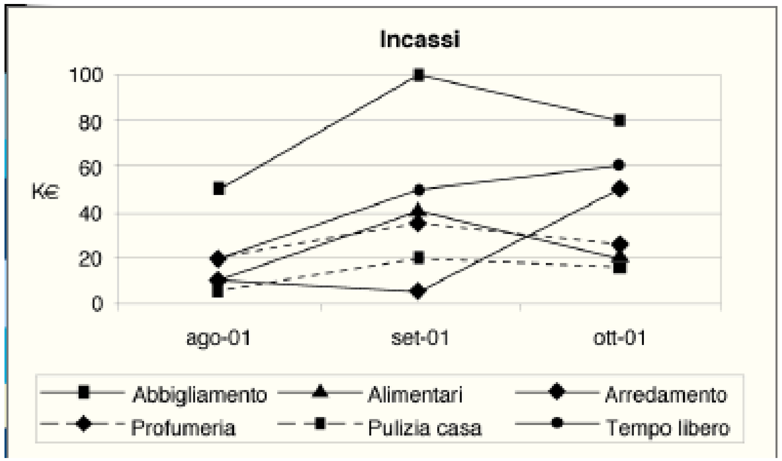
Filter Details:
Year = 1998

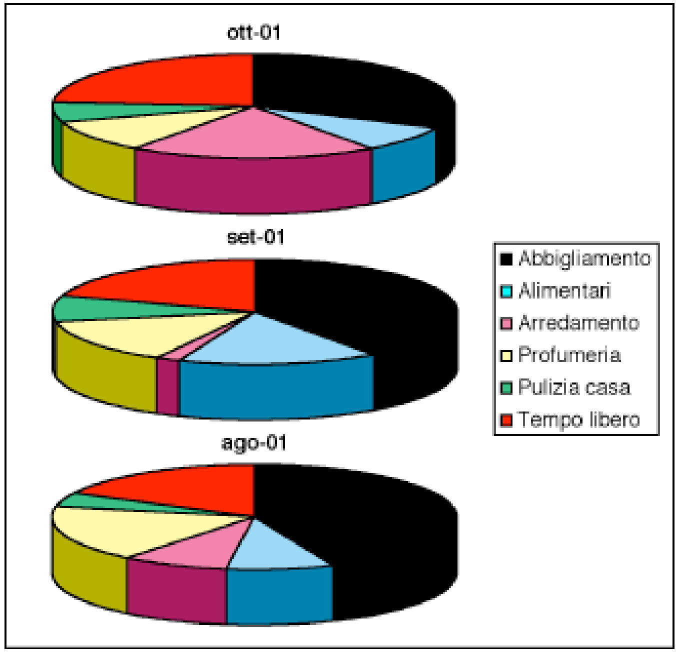
Category	Dollar Sales: North-East	Mid-Atlantic	South-East	Central	South	North-West	South-West	England
Electronics	$ 1.184	$ 4.529	$ 1.892	$ 7.232	$ 651	$ 9.488	$ 470	$ 2.083
Food	$ 538	$ 925	$ 959	$ 677	$ 213	$ 1.503	$ 261	$ 165
Gifts	$ 1.955	$ 2.785	$ 2.800	$ 2.695	$ 1.813	$ 2.844	$ 1.778	$ 1.158
Health & Beauty	$ 611	$ 887	$ 566	$ 382	$ 499	$ 1.162	$ 1.044	$ 273
Household	$ 5.787	$ 5.320	$ 5.416	$ 6.812	$ 4.334	$ 5.008	$ 7.588	$ 2.139
Kid's Korner	$ 247	$ 422	$ 441	$ 380	$ 221	$ 592	$ 290	$ 198
Travel	$ 608	$ 559	$ 1.096	$ 611	$ 464	$ 316	$ 573	$ 257

VISUALIZATION and REPORTS

Data may be visualized graphically, in an Excel-like format: tables, hystograms, graphics, 3D surfaces, etc.

incassi (K€)	Ottobre 2001	Settembre 2001	Agosto 2001
Abbigliamento	80	100	50
Alimentari	20	40	10
Arredamento	50	5	10
Profumeria	25	35	20
Pulizia casa	15	20	5
Tempo libero	60	50	20





Examples of operators: initial table

month	product	sales amount	quantity
February	pasta	130.000.000	45.000
March	pasta	140.000.000	50.000
April	pasta	135.000.000	51.000

Drill-down: adding a dimension

Drill-down on the zone

month	product	zone	quantity
February	pasta	north	15.000
February	pasta	east	17.000
February	pasta	center	13.000
March	pasta	north	18.000
March	pasta	east	18.000
March	pasta	center	14.000
April	pasta	north	18.000
April	pasta	east	17.000
April	pasta	center	16.000

Roll-up: dimension elimination

Roll-up on the month

product	zone	quantity
pasta	north	51.000
pasta	east	52.000
pasta	center	43.000

Aggregate Queries

Examples:

- Total sales per product category, per supermarket, per day
- Total monthly sales for all the products, per supermarket
- Total monthly sales per category per supermarket
- Avg. monthly sales per category, for all supermarkets

Olap logical models

- **MOLAP** (*Multidimensional On-Line Analytical Processing*) stores data by using a multidimensional data structure (a kind of “physical” data cube)

- **ROLAP** (*Relational On-Line Analytical Processing*) uses the relational data model to represent multidimensional data – <u>by far the most adopted solution</u>

The data cube: a new SQL command

- It expresses all the possible tuple aggregations of a table

- It uses the new polymorphic value **ALL** (in some systems **NULL** is used instead)

Example of Data Cube instruction in SQL (ROLAP)

```
select Model, Year,
       Color, sum(Sales)
from Sales
where Model in {'Fiat','Ford'}
  and Color = 'Red'
  and Year between 1994 and 1995
group by (Model, Year, Color)
with cube
```

Relevant Facts

model	year	color	sales
fiat	1994	red	50
fiat	1995	red	85
ford	1994	red	80

All the data in the cube

model	year	color	sum (sales)
fiat	1994	red	50
fiat	1995	red	85
fiat	1994	ALL	50
fiat	1995	ALL	85
fiat	ALL	red	135
fiat	ALL	ALL	135
ford	1994	red	80
ford	1994	ALL	80
ford	ALL	red	80
ford	ALL	ALL	80
ALL	1994	red	130
ALL	1995	red	85
ALL	ALL	red	215
ALL	1994	ALL	130
ALL	1995	ALL	85
ALL	ALL	ALL	215

Roll-up vs Data Cube

All the data in the cube

model	year	color	sum (sales)
fiat	1994	red	50
fiat	1995	red	85
fiat	1994	ALL	50
fiat	1995	ALL	85
fiat	ALL	red	135
fiat	ALL	ALL	135
ford	1994	red	80
ford	1994	ALL	80
ford	ALL	red	80
ford	ALL	ALL	80
ALL	1994	red	130
ALL	1995	red	85
ALL	ALL	red	215
ALL	1994	ALL	130
ALL	1995	ALL	85
ALL	ALL	ALL	215

CUBE evaluates aggregate expression with all possible combinations of columns specified in group by clause, whereas the Rollup evaluates aggregate expressions only relative to the order of columns specified in the group by clause.

```
select Model, Year,
       Color, sum(Sales)
from Sales
where Model in {'Fiat','Ford'}
  and Color = 'Red'
  and Year between 1994 and
 1995
group by (Model, Year, Color)
    with ROLLUP
```

The data after roll-up

model	year	color	sum(sales)
fiat	1994	red	50
fiat	1995	red	85
ford	1994	red	80
fiat	1994	ALL	50
fiat	1995	ALL	85
ford	1994	ALL	80
fiat	ALL	ALL	135
ford	ALL	ALL	80
ALL	ALL	ALL	215

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